water_use

Title

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Date: December 5, 2025

Regional Water Use Efficiency in California Vineyards

Author: William Mullins Github: https://github.com/willrmull/Vineyard_Water_Usage

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California agricultural water use accounts for approximately 80% of the state's developed water supply, with vineyards representing a significant portion of this usage. Thus, as increasing climate pressures begin to materialize it is important that we understand the regional differences in water use for sustainable resource management. This analysis examines the difference in applied water use across the hydrologic regions in California where vineyards are grown, controlling for climate factors, irrigation efficiency, and vineyard size to identify regions with superior water management practices.

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Figure 1: HR_ZONES

About the Data

The data used was collects from the California Department of Water Resources Excel Application Tool for **Statewide Agricultural Water Use Data from 2016 - 2020**. The data contains estimates for water use variables for twenty crop variables across the state. The variables used in this analysis are applied water (AW), irrigated crop area (ICA), crop evapotranspiration (ET_c), and effective precipitation (Ep) which were analysed at the hydrologic region level. The data and supporting metadata can be accessed here.

This Directed Acyclic Graph (DAG) illustrates the simplified relationships between the variables. 1. Hydrologic Region sits at the top as the primary driver of climate-influenced variables: - **Crop Evapotranspiration** (ET_c):The total water lost to the atmosphere from soil and plant surfaces. - **Effective Precipitation Volume**: he portion of rainfall that effectively contributes to meeting crop water needs.

2. Applied Water Volume (per acre): The "Applied Water Volume" for a single acre is determined in part by the difference between a crops water demand, ET_c, and the supply from effective precipitation—i.e., the deficit that must be supplied through irrigation.

3.Irrigated Crop Area acts as a scaling factor: total applied water volume is obtained by multiplying the per-acre applied water by the number of irrigated acres.

{ET_c}

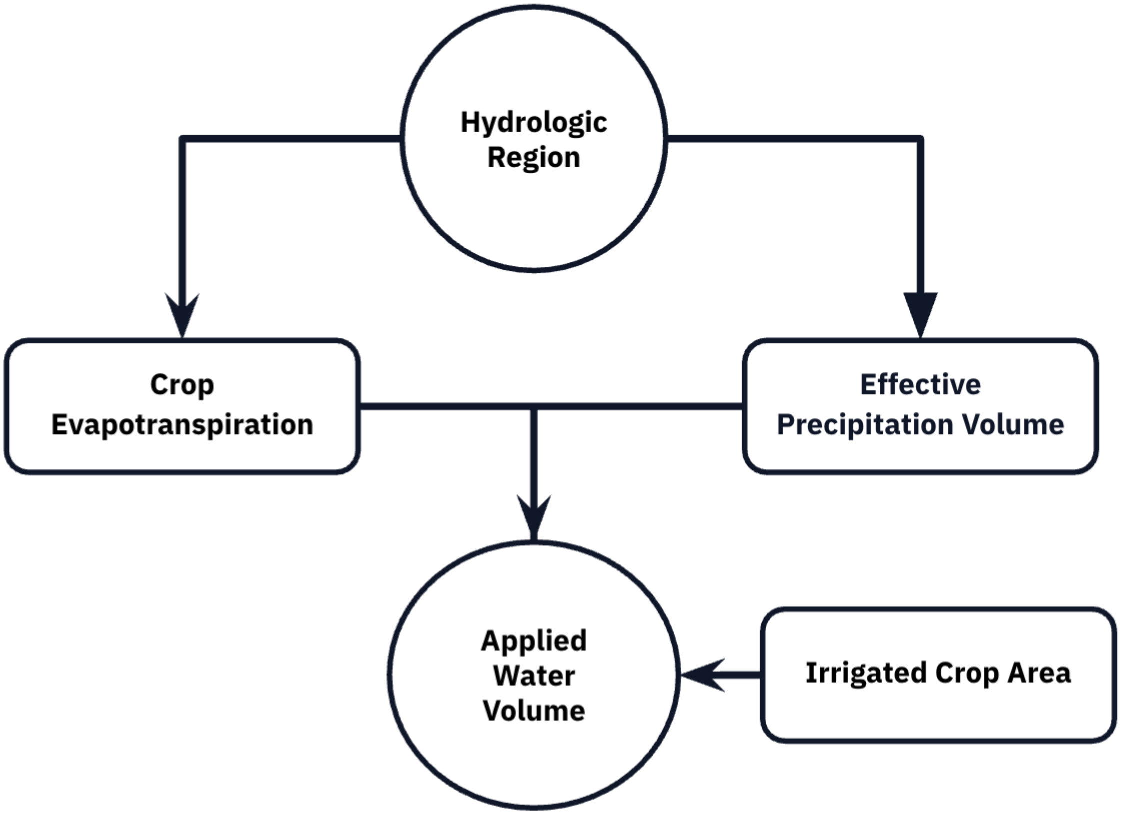


Figure 2: DAG

Import Packages

The analysis requires the following packages

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(viridis)
library(car)
library(patchwork)
library(broom)
```

```
Warning: package 'broom' was built under R version 4.5.2
```

```r
library(stringr)
```

Read in Data

```
New names:
New names:
New names:
New names:
* `Year` -> `Year...1`
* `RO` -> `RO...2`
* `HR` -> `HR...3`
* `PA` -> `PA...4`
* `DAU-Co` -> `DAU-Co...5`
* `Grain` -> `Grain...6`
* `Rice` -> `Rice...7`
* `Cotton` -> `Cotton...8`
* `Sugar Beet` -> `Sugar Beet...9`
* `Corn` -> `Corn...10`
* `Dry Beans` -> `Dry Beans...11`
* `Safflower` -> `Safflower...12`
* `Other Field` -> `Other Field...13`
* `Alfalfa` -> `Alfalfa...14`
* `Pasture` -> `Pasture...15`
* `Tomato Processing` -> `Tomato Processing...16`
* `Tomato Fresh` -> `Tomato Fresh...17`
* `Cucurbits` -> `Cucurbits...18`
* `Onion & Garlic` -> `Onion & Garlic...19`
* `Potatoes` -> `Potatoes...20`
* `Truck Crops` -> `Truck Crops...21`
* `Almonds & Pistachios` -> `Almonds & Pistachios...22`
* `Other Deciduous` -> `Other Deciduous...23`
* `Citrus & Subtropical` -> `Citrus & Subtropical...24`
* `Vineyard` -> `Vineyard...25`
* `` -> `...26`
* `Year` -> `Year...27`
* `RO` -> `RO...28`
* `HR` -> `HR...29`
```

* `PA` -> `PA...30`
* `DAUCo` -> `DAUCo...31`
* `Grain` -> `Grain...32`
* `Rice` -> `Rice...33`
* `Cotton` -> `Cotton...34`
* `Sugar Beets` -> `Sugar Beets...35`
* `Corn` -> `Corn...36`
* `Dry Beans` -> `Dry Beans...37`
* `Safflower` -> `Safflower...38`
* `Other Field` -> `Other Field...39`
* `Alfalfa` -> `Alfalfa...40`
* `Pasture` -> `Pasture...41`
* `Tomato Processing` -> `Tomato Processing...42`
* `Tomato Fresh` -> `Tomato Fresh...43`
* `Cucurbits` -> `Cucurbits...44`
* `Onion & Garlic` -> `Onion & Garlic...45`
* `Potatoes` -> `Potatoes...46`
* `Truck Crops` -> `Truck Crops...47`
* `Almonds & Pistachios` -> `Almonds & Pistachios...48`
* `Other Deciduous` -> `Other Deciduous...49`
* `Citrus & Subtropical` -> `Citrus & Subtropical...50`
* `Vineyard` -> `Vineyard...51`
* `` -> `...52`
* `Year` -> `Year...53`
* `RO` -> `RO...54`
* `HR` -> `HR...55`
* `PA` -> `PA...56`
* `DAU-Co` -> `DAU-Co...57`
* `Grain` -> `Grain...58`
* `Rice` -> `Rice...59`
* `Cotton` -> `Cotton...60`
* `Sugar Beet` -> `Sugar Beet...61`
* `Corn` -> `Corn...62`
* `Dry Beans` -> `Dry Beans...63`
* `Safflower` -> `Safflower...64`
* `Other Field` -> `Other Field...65`
* `Alfalfa` -> `Alfalfa...66`
* `Pasture` -> `Pasture...67`
* `Tomato Processing` -> `Tomato Processing...68`
* `Tomato Fresh` -> `Tomato Fresh...69`
* `Cucurbits` -> `Cucurbits...70`
* `Onion & Garlic` -> `Onion & Garlic...71`
* `Potatoes` -> `Potatoes...72`

* `Truck Crops` -> `Truck Crops...73`
* `Almonds & Pistachios` -> `Almonds & Pistachios...74`
* `Other Deciduous` -> `Other Deciduous...75`
* `Citrus & Subtropical` -> `Citrus & Subtropical...76`
* `Vineyard` -> `Vineyard...77`
* `` -> `...78`
* `Year` -> `Year...79`
* `RO` -> `RO...80`
* `HR` -> `HR...81`
* `PA` -> `PA...82`
* `DAUCo` -> `DAUCo...83`
* `Grain` -> `Grain...84`
* `Rice` -> `Rice...85`
* `Cotton` -> `Cotton...86`
* `Sugar Beet` -> `Sugar Beet...87`
* `Corn` -> `Corn...88`
* `Dry Beans` -> `Dry Beans...89`
* `Safflower` -> `Safflower...90`
* `Other Field` -> `Other Field...91`
* `Alfalfa` -> `Alfalfa...92`
* `Pasture` -> `Pasture...93`
* `Tomato Processing` -> `Tomato Processing...94`
* `Tomato Fresh` -> `Tomato Fresh...95`
* `Cucurbits` -> `Cucurbits...96`
* `Onion & Garlic` -> `Onion & Garlic...97`
* `Potatoes` -> `Potatoes...98`
* `Truck Crops` -> `Truck Crops...99`
* `Almonds & Pistachios` -> `Almonds & Pistachios...100`
* `Other Deciduous` -> `Other Deciduous...101`
* `Citrus & Subtropical` -> `Citrus & Subtropical...102`
* `Vineyard` -> `Vineyard...103`
* `` -> `...104`
* `Year` -> `Year...105`
* `RO` -> `RO...106`
* `HR` -> `HR...107`
* `PA` -> `PA...108`
* `DauCo` -> `DauCo...109`
* `Grain` -> `Grain...110`
* `Rice` -> `Rice...111`
* `Cotton` -> `Cotton...112`
* `Sugar Beets` -> `Sugar Beets...113`
* `Corn` -> `Corn...114`
* `Dry Beans` -> `Dry Beans...115`

* `Safflower` -> `Safflower...116`
* `Other Field` -> `Other Field...117`
* `Alfalfa` -> `Alfalfa...118`
* `Pasture` -> `Pasture...119`
* `Tomato Processing` -> `Tomato Processing...120`
* `Cucurbits` -> `Cucurbits...122`
* `Potatoes` -> `Potatoes...124`
* `Truck Crops` -> `Truck Crops...125`
* `Other Deciduous` -> `Other Deciduous...127`
* `Vineyard` -> `Vineyard...129`
* `Total AW` -> `Total AW...130`
* `` -> `...131`
* `Year` -> `Year...132`
* `RO` -> `RO...133`
* `HR` -> `HR...134`
* `PA` -> `PA...135`
* `DauCo` -> `DauCo...136`
* `Grain` -> `Grain...137`
* `Rice` -> `Rice...138`
* `Cotton` -> `Cotton...139`
* `Sugar Beets` -> `Sugar Beets...140`
* `Corn` -> `Corn...141`
* `Dry Beans` -> `Dry Beans...142`
* `Safflower` -> `Safflower...143`
* `Other Field Crops` -> `Other Field Crops...144`
* `Alfalfa` -> `Alfalfa...145`
* `Pasture` -> `Pasture...146`
* `Tomato Processing` -> `Tomato Processing...147`
* `Tomato Fresh` -> `Tomato Fresh...148`
* `Cucurbits` -> `Cucurbits...149`
* `Onions & Garlic` -> `Onions & Garlic...150`
* `Potatoes` -> `Potatoes...151`
* `Truck Crops` -> `Truck Crops...152`
* `Almonds & Pistachios` -> `Almonds & Pistachios...153`
* `Other Deciduous` -> `Other Deciduous...154`
* `Citrus & Subtropical` -> `Citrus & Subtropical...155`
* `Vineyard` -> `Vineyard...156`
* `Total AW` -> `Total AW...157`
* `` -> `...158`
* `Year` -> `Year...159`
* `RO` -> `RO...160`
* `HR` -> `HR...161`
* `PA` -> `PA...162`

* `DAUCo` -> `DAUCo...163`
* `Grain` -> `Grain...164`
* `Rice` -> `Rice...165`
* `Cotton` -> `Cotton...166`
* `Sugar Beets` -> `Sugar Beets...167`
* `Corn` -> `Corn...168`
* `Dry Beans` -> `Dry Beans...169`
* `Safflower` -> `Safflower...170`
* `Other Field` -> `Other Field...171`
* `Alfalfa` -> `Alfalfa...172`
* `Pasture` -> `Pasture...173`
* `Tomato Processing` -> `Tomato Processing...174`
* `Tomato Fresh` -> `Tomato Fresh...175`
* `Cucurbits` -> `Cucurbits...176`
* `Onion & Garlic` -> `Onion & Garlic...177`
* `Potatoes` -> `Potatoes...178`
* `Truck Crops` -> `Truck Crops...179`
* `Almonds & Pistachios` -> `Almonds & Pistachios...180`
* `Other Deciduous` -> `Other Deciduous...181`
* `Citrus & Subtropical` -> `Citrus & Subtropical...182`
* `Vineyard` -> `Vineyard...183`
* `Total AW` -> `Total AW...184`
* `` -> `...185`
* `Year` -> `Year...186`
* `RO` -> `RO...187`
* `HR` -> `HR...188`
* `PA` -> `PA...189`
* `DAU-Co` -> `DAU-Co...190`
* `Grain` -> `Grain...191`
* `Rice` -> `Rice...192`
* `Cotton` -> `Cotton...193`
* `Sugar Beets` -> `Sugar Beets...194`
* `Corn` -> `Corn...195`
* `Dry Beans` -> `Dry Beans...196`
* `Safflower` -> `Safflower...197`
* `Other Field` -> `Other Field...198`
* `Alfalfa` -> `Alfalfa...199`
* `Pasture` -> `Pasture...200`
* `Tomato Processing` -> `Tomato Processing...201`
* `Tomato Fresh` -> `Tomato Fresh...202`
* `Cucurbits` -> `Cucurbits...203`
* `Onion & Garlic` -> `Onion & Garlic...204`
* `Potatoes` -> `Potatoes...205`

```
* `Truck Crops` -> `Truck Crops...206`
* `Almonds & Pistachios` -> `Almonds & Pistachios...207`
* `Other Deciduous` -> `Other Deciduous...208`
* `Citrus & Subtropical` -> `Citrus & Subtropical...209`
* `Vineyard` -> `Vineyard...210`
* `Total AW` -> `Total AW...211`
* `` -> `...212`
* `Year` -> `Year...213`
* `RO` -> `RO...214`
* `HR` -> `HR...215`
* `PA` -> `PA...216`
* `DauCo` -> `DauCo...217`
* `Grain` -> `Grain...218`
* `Rice` -> `Rice...219`
* `Cotton` -> `Cotton...220`
* `Sugar Beets` -> `Sugar Beets...221`
* `Corn` -> `Corn...222`
* `Dry Beans` -> `Dry Beans...223`
* `Safflower` -> `Safflower...224`
* `Other Field Crops` -> `Other Field Crops...225`
* `Alfalfa` -> `Alfalfa...226`
* `Pasture` -> `Pasture...227`
* `Tomato Processing` -> `Tomato Processing...228`
* `Tomato Fresh` -> `Tomato Fresh...229`
* `Cucurbits` -> `Cucurbits...230`
* `Onions & Garlic` -> `Onions & Garlic...231`
* `Potatoes` -> `Potatoes...232`
* `Truck Crops` -> `Truck Crops...233`
* `Almonds & Pistachios` -> `Almonds & Pistachios...234`
* `Other Deciduous` -> `Other Deciduous...235`
* `Citrus & Subtropical` -> `Citrus & Subtropical...236`
* `Vineyard` -> `Vineyard...237`
* `Total AW` -> `Total AW...238`
```

Data Preperation

```r
# Select vineyard data
vineyard_data <- final_data %>%
  filter(Crop_Type == "Vineyard", Regional_AW_Vol > 0, Regional_ICA > 0) %>%
  mutate(
    across(Regional_AW_Vol:Regional_Ep_Vol, ~replace_na(., 0)), # Remove NA Variables
```

```
    aw_per_acre = Regional_AW_Vol / Regional_ICA,
    log_aw = log(Regional_AW_Vol)
  )

# Format Hydrologic Region Names
vineyard_data$HR <- gsub("0[0-9]_", "", vineyard_data$HR)
vineyard_data$HR <- gsub("10_", "", vineyard_data$HR)
vineyard_data$HR <- factor(vineyard_data$HR)

# View
head(vineyard_data)
```

```
# A tibble: 6 x 12
  Year  RO    HR            PA    `DAU-Co` Crop_Type Regional_AW_Vol Regional_ICA
  <chr> <chr> <fct>         <chr> <chr>    <chr>               <dbl>        <dbl>
1 2016  SCRO  Central Coa~ 301_~ 04827    Vineyard             2850.         1454
2 2016  SCRO  Central Coa~ 301_~ 04927    Vineyard             6422.         2987
3 2016  SCRO  Central Coa~ 301_~ 05027    Vineyard            46009.        20004
4 2016  SCRO  Central Coa~ 301_~ 05127    Vineyard            59265.        21551
5 2016  SCRO  Central Coa~ 301_~ 05327    Vineyard             1494.          661
6 2016  SCRO  Central Coa~ 301_~ 05427    Vineyard             3341.         1152
# i 4 more variables: Regional_ETc_Vol <dbl>, Regional_Ep_Vol <dbl>,
#   aw_per_acre <dbl>, log_aw <dbl>
```

Explore Data

Now that the data is clean we can gather summary statistics regarding the average annual water use across hydrologic regions which can be seen in the table below.

```
# Get Summary AW Data For Each Region
regional_summary <- vineyard_data %>%
  group_by(HR) %>%
  summarise(
    n = n(),
    mean_AW = mean(Regional_AW_Vol),
    median_AW = median(Regional_AW_Vol),
    sd_AW = sd(Regional_AW_Vol),
    se_AW = sd_AW / sqrt(n)) %>%
  arrange(mean_AW)
```

Applied Water (AW)

HR

n

mean_AW

median_AW

sd_AW

se_AW

South Lahontan

23

128.7751

39.4338

295.2821

61.57058

South Coast

46

1287.7218

296.4866

2120.1648

312.60125

Sacramento River

206

3631.8338

134.9146

8249.1706

574.74695

North Coast

121

5936.5745

443.8500

10137.3432

921.57665

Colorado River

25

6748.8049

386.1504

13446.1262

2689.22525

Central Coast

163

9917.9353

1476.9536

22978.5316

1799.81750

San Francisco Bay

80

10135.7991

138.5488

25803.3874

2884.90641

San Joaquin River

182

17764.8654

3262.9932

41878.1881

3104.21783

Tulare Lake

166

38072.2354

5663.5100

58246.3556

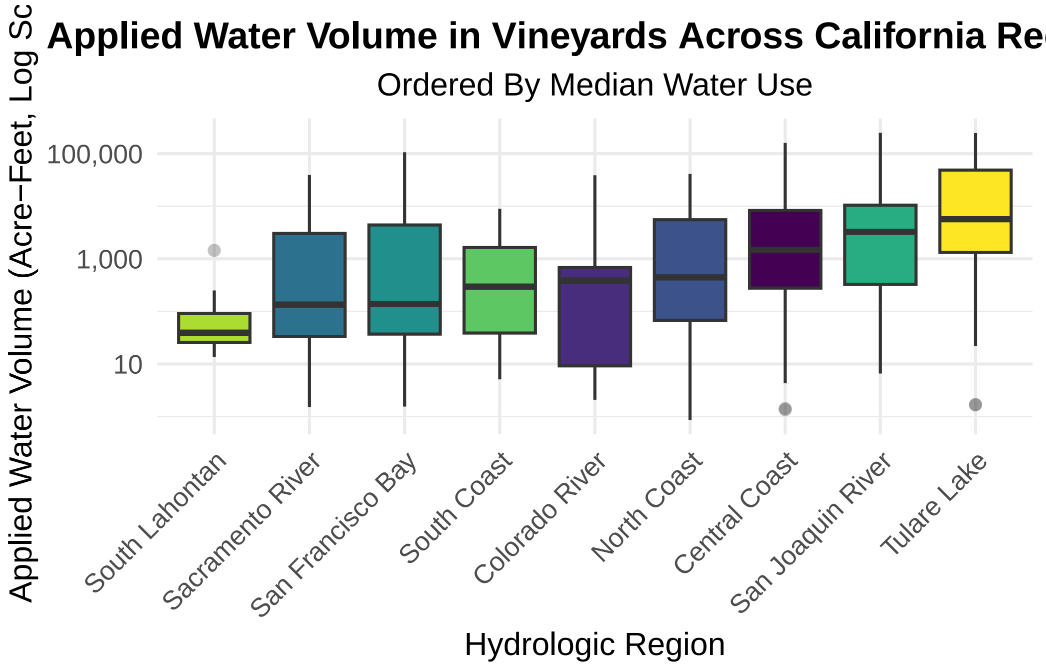
4520.79395

Takeaways:

- Mean AW Between Regions: significant variation in mean AW between regions, with it ranging from ~129 (South Lahontan) to ~38,072 (Tulare Lake).
- Variability Within Regions: In all of the regions the standard deviations exceeded the means, which indicates that there may be skewed distributions cause by high use vineyards
- Median < Mean: In all of the regions the median value was lower, often considerably so, than the mean indicating that there may be a right skew to the data.

This boxplot, which displays applied water volume on a log scale, helps visualize the significant variation within regions. It shows that most regions have modest median use, but a few (notably Central Coast and San Joaquin River) show extreme spread: from <10 to >100,000 acre-feet — driven by high-use outliers.

```
# Box-plot of Regional_AW_Vol
vineyard_data %>%
  ggplot(aes(x = reorder(HR, Regional_AW_Vol, FUN = median),  # Order Data By Median
             y = Regional_AW_Vol)) +
  geom_boxplot(aes(fill = HR), outlier.alpha = 0.3) +
  scale_fill_viridis_d() +
  scale_y_log10(labels = scales::comma) + # Apply log scale
  labs(
    x = "Hydrologic Region",
    y = "Applied Water Volume (Acre-Feet, Log Scale)",
    title = "Applied Water Volume in Vineyards Across California Regions",
    subtitle = "Ordered By Median Water Use"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none",
    plot.title = element_text(hjust = 0.5, face = "bold"),
    plot.subtitle = element_text(hjust = 0.5)
  )
```



Gamma Model

For this analysis a gamma model will be used, which shows the.

Applied Water Volume ~ Gamma(,)

log( ) = 0 + 1$ETc\ Volume$ + 2Ep Volume + 3$log(ICA)$ + 4HR

The Gamma GLM with log link was used for the following reasons: - Positive Values Only: The water use can never be zero or negative in areas which are irrigated - Right Skewed: As mentioned previously the data shows an extreme positive skews in water use

```
model_gamma <- glm(
  Regional_AW_Vol ~
    Regional_ETc_Vol +
    Regional_Ep_Vol +
    log(Regional_ICA) +
    HR,
  family = Gamma(link = "log"),
  data = vineyard_data
)

summary(model_gamma)
```

```
Call:
glm(formula = Regional_AW_Vol ~ Regional_ETc_Vol + Regional_Ep_Vol +
    log(Regional_ICA) + HR, family = Gamma(link = "log"), data = vineyard_data)

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          7.473e-01  1.753e-02  42.624  < 2e-16 ***
Regional_ETc_Vol     9.758e-07  3.035e-07   3.215  0.00135 **
Regional_Ep_Vol     -7.294e-06  2.672e-06  -2.730  0.00645 **
log(Regional_ICA)    1.010e+00  2.213e-03 456.617  < 2e-16 ***
HRColorado River     6.531e-01  3.313e-02  19.712  < 2e-16 ***
HRNorth Coast       -1.257e-02  1.857e-02  -0.677  0.49850
HRSacramento River   2.045e-01  1.645e-02  12.432  < 2e-16 ***
HRSan Francisco Bay  5.207e-02  2.117e-02   2.460  0.01406 *
HRSan Joaquin River  2.495e-01  1.660e-02  15.029  < 2e-16 ***
HRSouth Coast        3.332e-01  2.572e-02  12.956  < 2e-16 ***
HRSouth Lahontan     5.557e-01  3.475e-02  15.993  < 2e-16 ***
HRTulare Lake        5.399e-01  1.755e-02  30.769  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.02336771)

    Null deviance: 5742.400  on 1011  degrees of freedom
Residual deviance:   23.979  on 1000  degrees of freedom
AIC: 12624

Number of Fisher Scoring iterations: 4
```

Hypothesis Test: No Regional Differences

Null hypothesis: No difference in water use between regions (all HR coefficients = 0)
Alternative hypothesis: Water use varies in at least one of the regions.

This null hypothesis is tested by comparing the original full model with one without regions. By including the other vartiables in bo

```r
# Null model (without HR)
model_null <- glm(
  Regional_AW_Vol ~
    Regional_ETc_Vol +
```

```r
    Regional_Ep_Vol +
    log(Regional_ICA),
  family = Gamma(link = "log"),
  data = vineyard_data
)

lrt <- anova(model_null, model_gamma, test = "LRT")
print(lrt)
```

```
Analysis of Deviance Table

Model 1: Regional_AW_Vol ~ Regional_ETc_Vol + Regional_Ep_Vol + log(Regional_ICA)
Model 2: Regional_AW_Vol ~ Regional_ETc_Vol + Regional_Ep_Vol + log(Regional_ICA) +
    HR
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      1008     63.545
2      1000     23.979  8   39.566 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
print(paste("Chi-square statistic:", lrt$Deviance[2]))
```

```
[1] "Chi-square statistic: 39.5657954632089"
```

```r
print(paste("Degrees of freedom:", lrt$Df[2]))
```

```
[1] "Degrees of freedom: 8"
```

```r
print(paste("P-value:", lrt$`Pr(>Chi)`[2]))
```

```
[1] "P-value: 0"
```

Interpretation of Model Results

Chi-square statistic: 39.5658 Degrees of freedom: 8 P-value: < 0.001 (1.67×10^{2})

We reject the null hypothesis. There are statistically significant differences in applied water volume (Regional_AW_Vol) between hydrologic regions (HR), even after controlling for: - Crop water demand (Regional_ETc_Vol) - Natural water supply (Regional_Ep_Vol) - Vineyard size (log(Regional_ICA)).

TYPE III ANOVA: INDIVIDUAL PREDICTOR SIGNIFICANCE

```r
anova_results <- Anova(model_gamma, type = "III", test.statistic = "Wald")
print(anova_results)
```

```
Analysis of Deviance Table (Type III tests)

Response: Regional_AW_Vol
                  Df      Chisq Pr(>Chisq)
(Intercept)        1 1.8168e+03  < 2.2e-16 ***
Regional_ETc_Vol   1 1.0336e+01   0.001305 **
Regional_Ep_Vol    1 7.4503e+00   0.006343 **
log(Regional_ICA)  1 2.0850e+05  < 2.2e-16 ***
HR                 8 1.6302e+03  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Parameter Signifigance

Hydrologic Region - Strongest regional effect (^{2} = 1,630.2, $p < 0.001$). - Interpretation: Regional policies, infrastructure, soil types, or unmeasured climatic factors significantly drive water use differences beyond climate and vineyard size.

Vineyard Size (log(Regional_ICA)) - Extremely significant (^{2} = 208,500, $p < 0.001$).

Climate Drivers - Crop ET (Regional_ETc_Vol): Significant ($p = 0.0013$). Higher water demand $\rightarrow$ more applied water. - Effective Precip (Regional_Ep_Vol): Significant ($p = 0.0063$). Natural rainfall offsets irrigation needs.

Intercept - Baseline water use (when all predictors = 0) is non-zero ($p < 0.001$), reflecting fixed irrigation requirements.

Parameter Simulation

```r
# Get coefficient values and dispersion from gamma model
true_params <- coef(model_gamma)
true_dispersion <- summary(model_gamma)$dispersion
```

```r
# function that generates synthetic data
simulate_vineyard_data <- function(n_obs = 1000) {

  # Samples 1,000 observations with replacement from your original dataset
  sim_data <- vineyard_data %>%
    sample_n(n_obs, replace = TRUE) %>%
    select(Regional_ETc_Vol, Regional_Ep_Vol, Regional_ICA, HR)

  # Calculate Expected Values
  X <- model.matrix(~ Regional_ETc_Vol + Regional_Ep_Vol +
                    log(Regional_ICA) + HR, data = sim_data)
  # Calculates the linear predictor
  eta <- X %*% true_params
  # inverse log link function to get expected water use values
  mu <- exp(eta)

  # Simulate Response Variables
  shape_param <- 1 / true_dispersion
  sim_data$Regional_AW_Vol <- rgamma(
    n_obs,
    shape = shape_param,
    scale = as.vector(mu) * true_dispersion
  )

  return(sim_data)
}

# Run simulation
set.seed(123)
vineyard_sim <- simulate_vineyard_data(n_obs = 1000)

# Refit model to simulated data
validation_model <- glm(
  Regional_AW_Vol ~ Regional_ETc_Vol + Regional_Ep_Vol +
    log(Regional_ICA) + HR,
  family = Gamma(link = "log"),
  data = vineyard_sim
)
```

```r
# Compare parameters from gamma model and validation model
param_comparison <- data.frame(
  Parameter = names(true_params),
```

```r
  True = true_params,
  Recovered = coef(validation_model),
  Difference = true_params - coef(validation_model),
  Pct_Error = abs((true_params - coef(validation_model)) / true_params * 100)
)

print(param_comparison)
```

```
                              Parameter          True     Recovered
(Intercept)                 (Intercept)  7.472619e-01  7.426626e-01
Regional_ETc_Vol       Regional_ETc_Vol  9.757792e-07  8.633499e-07
Regional_Ep_Vol         Regional_Ep_Vol -7.293923e-06 -8.424088e-06
log(Regional_ICA)     log(Regional_ICA)  1.010415e+00  1.010039e+00
HRColorado River       HRColorado River  6.531031e-01  6.622381e-01
HRNorth Coast             HRNorth Coast -1.257058e-02 -5.218514e-03
HRSacramento River   HRSacramento River  2.045434e-01  1.988061e-01
HRSan Francisco Bay HRSan Francisco Bay  5.207079e-02  3.965745e-02
HRSan Joaquin River HRSan Joaquin River  2.495017e-01  2.610238e-01
HRSouth Coast             HRSouth Coast  3.332208e-01  3.451779e-01
HRSouth Lahontan       HRSouth Lahontan  5.557495e-01  5.572932e-01
HRTulare Lake             HRTulare Lake  5.398885e-01  5.556828e-01
                       Difference  Pct_Error
(Intercept)          4.599331e-03  0.6154912
Regional_ETc_Vol     1.124293e-07 11.5220036
Regional_Ep_Vol      1.130165e-06 15.4946059
log(Regional_ICA)    3.759793e-04  0.0372104
HRColorado River    -9.135027e-03  1.3987114
HRNorth Coast       -7.352071e-03 58.4863066
HRSacramento River   5.737284e-03  2.8049231
HRSan Francisco Bay  1.241334e-02 23.8393504
HRSan Joaquin River -1.152210e-02  4.6180440
HRSouth Coast       -1.195710e-02  3.5883414
HRSouth Lahontan    -1.543710e-03  0.2777709
HRTulare Lake       -1.579427e-02  2.9254690
```

```r
print(paste("Mean absolute % error:", round(mean(param_comparison$Pct_Error, na.rm = TRUE), 2
```

```
[1] "Mean absolute % error: 10.47"
```

```r
print(paste("Max absolute % error:", round(max(param_comparison$Pct_Error, na.rm = TRUE), 2)
```

```
[1] "Max absolute % error: 58.49"
```

Parameter Recovery Summary: - Mean absolute % error: 10.47 - Max absolute % error: 58.49"

These values indicate that the simulation is not recovering the parameters well.This indicates that the model can reliably estimate recovery coefficients.

Visualization of Model Fit

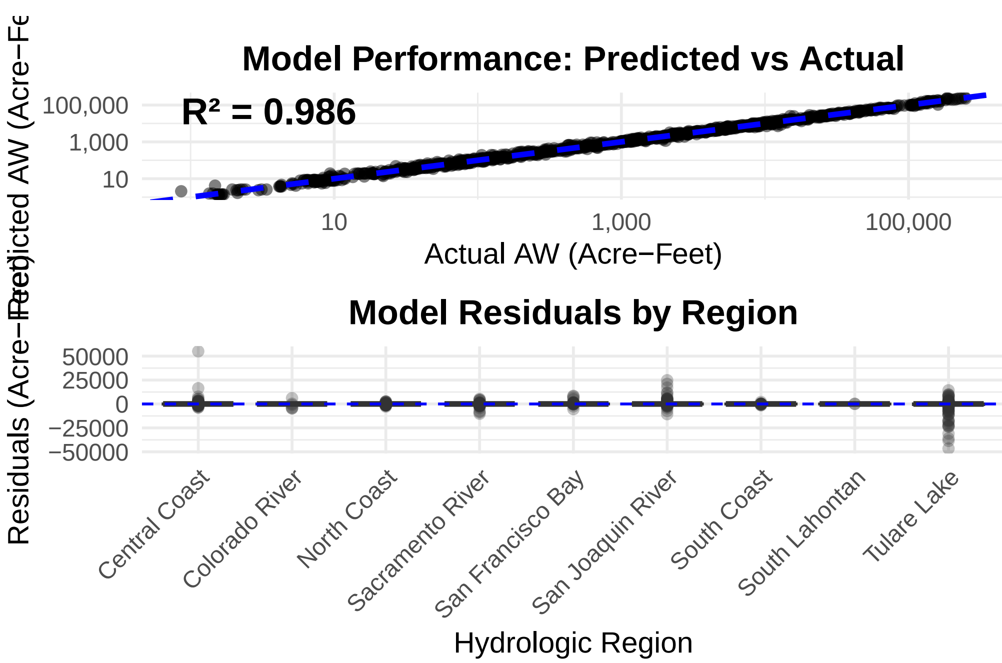
```r
# Get predictions
predicted <- predict(model_gamma, type = "response")
comparison <- data.frame(
  Actual = vineyard_data$Regional_AW_Vol,
  Predicted = predicted,
  Residual = residuals(model_gamma, type = "response"),
  HR = vineyard_data$HR
)
r_squared <- cor(comparison$Actual, comparison$Predicted)^2

# Predicted vs Actual
predicted_plot <- ggplot(comparison, aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.5) +
  geom_abline(slope = 1, intercept = 0, color = "blue",
              linetype = "dashed", linewidth = 1) +
  scale_x_log10(labels = scales::comma) +
  scale_y_log10(labels = scales::comma) +
  scale_color_viridis_d() +
  annotate("text", x = min(comparison$Actual), y = max(comparison$Predicted),
           label = paste0("R² = ", round(r_squared, 3)),
           hjust = 0, vjust = 1, size = 5, fontface = "bold") +
  labs(
    title = "Model Performance: Predicted vs Actual",
    x = "Actual AW (Acre-Feet)",
    y = "Predicted AW (Acre-Feet)"
  ) +
  theme_minimal(base_size = 11) +
  theme(
```

```
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.position = "right"
  )

# Residuals by region
residutal_plot <- ggplot(comparison, aes(x = HR, y = Residual)) +
  geom_boxplot(aes(fill = HR), outlier.alpha = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "blue") +
  scale_fill_viridis_d() +
  labs(
    title = "Model Residuals by Region",
    x = "Hydrologic Region",
    y = "Residuals (Acre-Feet)"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none",
    plot.title = element_text(hjust = 0.5, face = "bold")
  )

print(predicted_plot / residutal_plot)
```



Discussion

Regional Differences in Water Use Efficiency

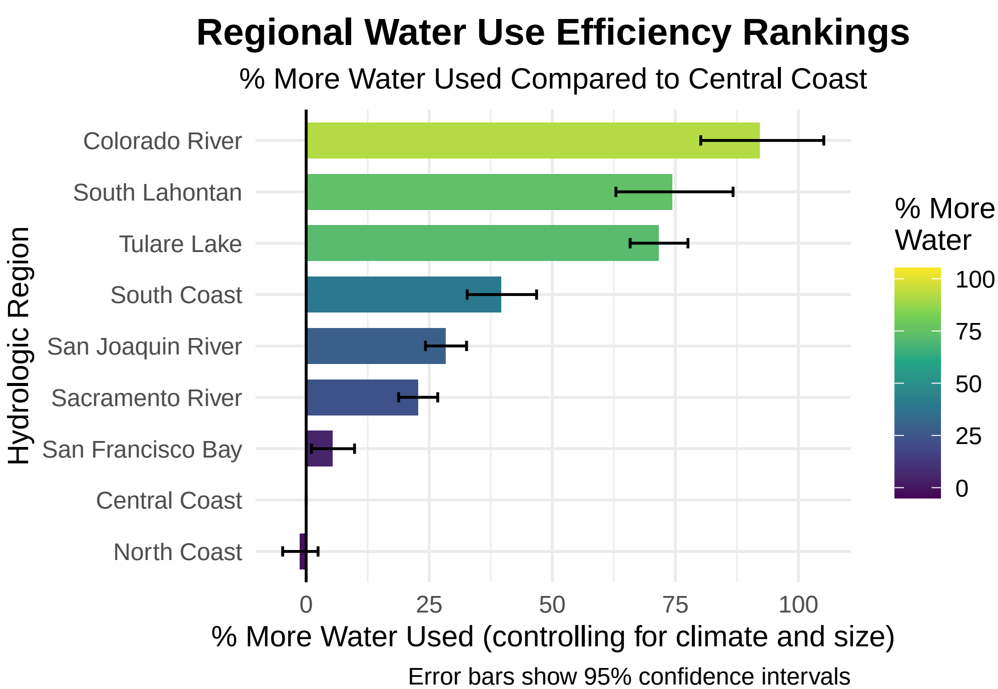
Add Explanation

```r
# Extract regional coefficients
regional_coefs <- tidy(model_gamma, conf.int = TRUE) %>%
  filter(str_starts(term, "HR")) %>%
  # Converts log-scale estimates to percentage differences in water use
  mutate(
    region = str_remove(term, "^HR"),
    pct_more_water = (exp(estimate) - 1) * 100,
    ci_low_pct = (exp(conf.low) - 1) * 100,
    ci_high_pct = (exp(conf.high) - 1) * 100
  ) %>%
  # Sets values from reference region (Central Coast) to zero
  bind_rows(
    tibble(
      term = "HR(Reference)",
      estimate = 0,
      std.error = 0,
      statistic = 0,
      p.value = NA,
      conf.low = 0,
      conf.high = 0,
      region = levels(vineyard_data$HR)[1],
      pct_more_water = 0,
      ci_low_pct = 0,
      ci_high_pct = 0
    )
  ) %>%
  arrange(desc(estimate))  # Fixed ranking direction

# Visualize
plot_rank <- ggplot(regional_coefs,  aes(x = reorder(region, estimate), y = pct_more_water))
  geom_col(aes(fill = pct_more_water), width = 0.7) +
  geom_errorbar(aes(ymin = ci_low_pct, ymax = ci_high_pct), width = 0.2) +
  geom_hline(yintercept = 0) +
   scale_fill_viridis_c(
    direction = 1,
    name = "% More\nWater",
```

```
    limits = c(min(regional_coefs$ci_low_pct), max(regional_coefs$ci_high_pct))
  ) +
  coord_flip(clip = "off") +
  labs(
    title = "Regional Water Use Efficiency Rankings",
    subtitle = "% More Water Used Compared to Central Coast",
    x = "Hydrologic Region",
    y = "% More Water Used (controlling for climate and size)",
    caption = "Error bars show 95% confidence intervals"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
    plot.subtitle = element_text(hjust = 0.5),
    legend.position = "right",
    plot.margin = margin(r = 20)  # Extra space for rank numbers
  )

print(plot_rank)
```



Add Stats Later

Discussion: